# RADIOSONDE TEST SET TS-1348( )/GMM-1

## 1. SCOPE

1.1 Scope.- This specification covers one type of equipment designated as Radiosonde Test Set TS-1348( )/GMM-1. (See 6.1 and 6.4).

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, or request for proposal form a part of this specification to the extent stated herein.

### SPECIFICATIONS

#### FEDERAL

| | |
|---|---|
| L-V-351 | Vinyl Chloride Polymer and Copolymer Rigid Sheets |
| PPP-B-585 | Boxes, Wood, Wirebound |
| PPP-B-591 | Boxes, Fiberboard, Wood-Cleated |
| PPP-B-601 | Boxes, Wood, Cleated-Plywood |
| PPP-B-621 | Boxes, Wood, Nailed and Lock-Corner |
| PPP-B-636 | Boxes, Fiber |
| PPP-T-97 | Tape; Pressure-Sensitive Adhesive, Paper, Water-Resistant |

#### MILITARY

| | |
|---|---|
| MIL-R-26 | Resistors, Fixed, Wire-Wound, (Power Type) |
| MIL-T-27 | Transformers and Inductors, (Audio, Power, and Pulse) |
| MIL-R-94 | Resistors, Variable, Composition |
| MIL-P-116 | Preservation Methods of |
| MIL-C-3442 | Cable and Wire, Electrical |
| MIL-S-3786 | Switches, Rotary, (Circuit Selector, Low Current Capacity) |
| MIL-C-3885 | Cable Assemblies & Cord Assemblies, Electrical |



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MILITARY (Cont'd)

| | |
|---|---|
| MIL-K-3926 | Knobs, Control ( For use with Electronic, Communications, and Allied Equipment) |
| MIL-C-5015 | Connectors, Electric, AN Type |
| MIL-B-10154 | Batteries, Primary, Water-Activated (Dunk Type) |
| MIL-M-10304 | Meters, Electrical Indicating, Panel Type, Ruggedized |
| MIL-P-11268 | Parts, Materials, and Processes |
| MIL-M-13231 | Marking of Electronic Items |
| MIL-F-14072 | Finishes for Ground Signal Equipment |
| MIL-F-15160 | Fuses; Instrument, Power, and Telephone |
| MIL-S-19500 | Semiconductor Devices |
| MIL-M-19833 | Molding Plastic and Molded Plastic Parts, (Glass-Fiber-Filled, Diallyl-Phthalate Resin) |
| MIL-C-55169 | Connectors, Electric, Type "Q", Multi-contact. |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-169 | Extreme Temperature Cycle |
| MIL-STD-170 | Moisture Resistance Test Cycle for Ground Signal Equipment |
| MIL-STD-202 | Test Methods for Electronic and Electrical Component Parts |
| MIL-STD-252 | Wired Equipment, Classification of Visual and Mechanical Defects |
| MS 90120 | Knobs, Control, Bar, Plastic |
| MS 91528 | Knobs, Control, Round, Plastic (with ribbed gripping surfaces) |

DRAWINGS

SIGNAL CORPS

| | |
|---|---|
| SC-A-46439 | List of Accessories for Package Tester |
| SC-A-362100 | Requirements for Preparation of Packaging and Packing Illustrations and Bills of Material |

(Copies of specifications, standards, and drawings, by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)



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## 3. REQUIREMENTS

3.1 Description.- Radiosonde Test Set TS-1348( )/GMM-1 is a small unit incorporating voltmeters and ammeters, together with auxilliary items, inclosed in a metal case, which is fastened to the top of Radiosonde Baseline Check Set AN/GMM-1.

3.2 Procurement model.- A procurement model of the unit will be available for inspection by prospective bidders and will be lent to the contractor. Unless otherwise specified herein, or in the invitation for bids, physical construction of the unit shall conform to the model. (NOTE: In case of conflict between specified performance characteristics for the unit and the performance of the model, the specified performance characteristics govern.)

3.3 Preproduction samples.- The contractor shall furnish two preproduction samples for inspection and approval. (See 4.3).

3.4 Parts, materials, and processes.- Unless otherwise specified, and in addition to the requirements of this specification, the Specification MIL-P-11268, including the selection requirements therein, shall apply.

3.4.1 Transformer.- The power transformer shall be Grade 4, Class R, per Specification MIL-T-27.

3.4.2 Converter.- The D.C. to D.C. converter shall deliver 34 milliamperes at 115 ± 5 volts with an input of 24 volts. This unit shall be hermetically sealed, except for the transistors, which shall be accessible for replacement purposes.

3.4.2.1 Transient protection.- The unit shall be so designed that the wave shape, peak to peak amplitude, including any transients between the collector and emitter of the transistors shall not exceed the rated collector to [illegible]er breakdown voltage.

3.4.2.2 Regulation.- The variation in output voltage from 50 to 100 percent of full load shall be between 97 and 100 percent of the rated full load voltage.

3.4.2.3 Output ripple.- The output ripple shall not exceed 0.5 volt, r.m.s., at a nominal input of 24 volts.

3.4.3 Meters.- The meters in the set shall conform to Specification MIL-M-10304 and be of the following types:

1 MR 26W 50DCMAR
2 MR 26W 3DCVVR
3 MR 26W 10DCVVR
4 MR 26W 150DCVVR

3.4.3.1 Milliammeter.- The milliammeter shall be shunted and wired to the switch to provide full-scale-reading of 250 m.a.

3.4.4 Connector.- The radiosonde chassis connector insert shall conform to Specification MIL-M-19833, and consist of glass-fibre-filled dially-pthalate, Grade GD 1. The pin terminals on connectors shall conform to Specification MIL-C-5015.



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3.4.5 Cable assemblies.- Cable shall conform to Specification MIL-C-3442 and cable assemblies shall conform to Specification MIL-C-3885.

3.4.5.1 Power cable.- The power cable shall be type CO-02HGF (2/18) 0390. The length shall be 15 feet, -0, + 3 inches. Terminals shall be as follows:

Male Twist Lock (Hubbell, Type 7429, or equal)
Connector, Q-type, Q2A12A3P, per Specification MIL-C-55169, including Gland & Washer assembly

3.4.5.2 Output cable.- The output cable shall be Type CO-05MGE(5/20)0400. The length shall be 3 feet -0 + 1-3/8 inches. Terminals shall be as follows:

Male Standard AN, 5-pin connector
Female - (conforming to the socket on the Battery Pack BA-259/AM, per Specification MIL-B-10154.

3.4.6 Resistors.-

3.4.6.1 Power resistor.- The wire-wound power resistor shall conform to Specification MIL-R-26 and shall be marked with the type designation.

3.4.6.2 Load resistors.- Simulated load resistors shall be provided having values as follows:

$A_1$ Load: 12 ohms A Load: 36 ohms B Load: 3300 ohms

Resistance tolerances are 5% for each circuit. The resistor in the $A_1$ load circuit shall have a value such that together with the resistance of the fuse, the total resistance will be as specified.

3.4.6.3 Voltage divider.- Resistive voltage dividers shall be used to supply the $A_1$ and A voltages directly when a 24 volt direct current power source is employed.

3.4.7 Switches.- Wafer type switches shall conform to Specification MIL-S-3786, and shall provide the following functions:

3.4.7.1 Load selector.- To apply the operating voltage to either the Radiosonde or to the Dummy Load.

3.4.7.2 Load current selector.- To provide current measurements in any one of the three load circuits while retaining continuity of the other load circuits meanwhile.

3.4.7.3 Power selector.- To obtain power from either the battery or from the power unit in the set. This switch shall also have an OFF position so that all power can be taken off the loads.



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3.4.8 Potentiometers.- Potentiometers shall conform to Specification MIL-R-94 and shall provide for adjustment of each of the several voltages metered.

3.4.9 Fuses.- Fuses shall conform to Specification MIL-F-15160 and shall serve to protect each of the load circuits. Access shall be from the front panel.

3.4.10 Knobs.- Knobs shall conform to Specification MIL-K-3926. Switches shall be fitted with Part Number MS-90120-2801; Potentiometers shall be fitted with Part Number MS-91528-1D28.

3.4.11 Overall dimensions and weight.- The overall dimensions of the unit shall not exceed the following:

| | | |
|---|---|---|
| Length | - | 17 inches |
| Depth | - | 6 inches |
| Height | - | 4 inches |

The weight shall not exceed ten pounds.

3.4.12 Transistors.- Transistors shall conform to MIL-S-19500.

3.5 Marking.- Marking shall conform to the requirements of Group I in Specification MIL-M-13231.

3.5.1 Symbols.- Reference symbols for component parts shall conform to Specification MIL-M-13231.

3.5.2 Instruction plate.- Instructions, briefly stated, pertaining to the use of the unit shall be affixed to the top of the unit. The instructions shall be printed using black letters on a white background, or green letters on a light yellow background. In either case the lettering shall be Gothic lightface type not smaller than 6 point, number 4. The instructions shall be laminated within vinyl-chloride sheet, Class 1, conforming to Federal Specification L-V-351, and firmly attached to the unit. Specific approval of the instruction plate shall be obtained before the production is begun.

3.5.3 Panel layout and marking.- Marking shall conform to the requirements of Group 1 of Specification MIL-M-13231.

3.5.4 Serial numbers.- Serial numbers shall be affixed on all production units.

3.6 Finish.- The unit shall be finished in accordance with Specification MIL-F-14072. This includes finish of hardware such as clamps, hinges, and screws, including necessary touch-up after mounting. The final paint film on type 1 surfaces shall be green in color (olive drab), semi-gloss enamel, matching the color chip No. X-24087 supplied by the contracting officer. (See 6.3.)



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3.7 Performance.- The electrical performance of the unit shall conform to the following requirements:

3.7.1 Insulation.- Insulation shall withstand the breakdown test of 4.9.2.

3.7.2 Power supply.- The unit shall supply power in accordance with Table I when connected to a 24 volt D.C. power source such as a battery on a vehicular transport unit.

Table I.- Power Requirements

| Power Level | Voltage Range | Current in Milliamperes |
|---|---|---|
| $A_1$ | 1.1 to 1.5 | 110 at 1.4 volts |
| A | 5.4 to 7.4 | 180 at 6.6 volts |
| B | 95 to 120 | 34 at 115 volts |

3.7.3 Current and voltage.- When using the simulated loads and when supplied with the voltages specified in Table VIII, the performance, in terms of the several meter readings, shall conform to the values in Table II, for the switch positions therein indicated.



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Table II.- Performance

| Switch Positions | | | Voltmeter Readings | | | Load Current in Milliamperes |
|---|---|---|---|---|---|---|
| Load Selector | Load Current Selector | Power Selector | $A_1$ | A | B | |
| Radiosonde | $A_1$ | 24 vdc | 1.1 ± 16% | 4.4 ± 16% | 115 ± 4% | 94 ± 10% |
| | A | 24 vdc | 1.1 ± 16% | 4.4 ± 16% | 115 ± 4% | 134 ± 9% |
| | B | 24 vdc | 1.1 ± 16% | 4.4 ± 16% | 115 ± 4% | 21 ± 11% |
| Radiosonde | $A_1$ | Battery | 1.4 ± 6% | 6.6 ± 4% | 115 ± 4% | 25 ± 22% |
| | A | Battery | 1.4 ± 6% | 6.6 ± 4% | 115 ± 4% | 81 ± 11% |
| | B | Battery | 1.4 ± 6% | 6.6 ± 4% | 115 ± 4% | 21 ± 11% |
| Dummy | $A_1$ | 24 vdc | 1.4 ± 6% | 6.6 ± 4% | 115 ± 4% | 116 ± 9% |
| | A | 24 vdc | 1.4 ± 6% | 6.6 ± 4% | 115 ± 4% | 183 ± 8% |
| | B | 24 vdc | 1.4 ± 6% | 6.6 ± 4% | 115 ± 4% | 35 ± 9% |
| Dummy | $A_1$ | Battery | 1.4 ± 6% | 6.6 ± 4% | 115 ± 4% | 94 ± 9% |
| | A | Battery | 1.4 ± 6% | 6.6 ± 4% | 115 ± 4% | 183 ± 8% |
| | B | Battery | 1.4 ± 6% | 6.6 ± 4% | 115 ± 4% | 35 ± 9% |

3.8 Preconditioning.- The unit shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.7.

3.9 Service conditions.- The unit shall withstand the following service conditions without impairment of the performance required in 3.7.

3.9.1 Temperature.-

3.9.1.1 Operating.- Ambient temperature in the range of -40 to + 150 degrees Fahrenheit. (The 150 degree temperature includes the effect of run-load.) Exposure at the high temperature not to exceed four hours, and at the low temperature, not to exceed 72 hours, at any one time.

3.9.1.2 Non-operating.- Exposure in the range of - 80 to + 160 degrees F. Exposure at the high extreme not to exceed four hours, and at the low temperature not to exceed 24 hours, at any one time.

3.9.2 Altitude.-

3.9.2.1 Operating.- Up to 10,000 feet above sea level.



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3.9.2.2 Non-operating.- Up to 50,000 feet above sea level.

3.9.3 Moisture-resistance (relative humidity).- Up to 97 percent relative humidity for a period of 20 hours, and exposure at 100 percent relative humidity, with condensation, for four hours.

3.9.4 Rain.- Two inches of rain per hour, with wind at 20 miles an hour.

3.9.5 Sand and dust.- As encountered over dry clay road, or in desert areas, during heavy vehicular or personnel traffic.

3.9.6 Vibration.- The amplitude of any part, subassembly, or structural member, of the unit shall not exceed twice the amplitude of the vibration applied to the unit at any frequency between 10 and 55 cycles per second.

3.9.7 Bounce.- The unit shall withstand the test in 4.17 without damage and with no impairment of performance, except minor surface abrasion.

3.9.8 Shock.- The unit shall withstand the test in 4.18 without damage and with no impairment of performance.

3.10 Technical literature, and running spares.- Technical literature and running spares shall be provided as specified in the contract. Spares shall be identical to corresponding parts for the unit supplied on the order.

3.11 Workmanship.- Workmanship shall meet the requirements of this specification and of the specifications cited herein subsidiary thereto. (See 4.4.1). The unit shall be manufactured and assembled in accordance with the applicable portions of the following requirements in Specification MIL-P-11268:

Wiring and cabling, including all subsidiary paragraphs.
Flux, cleaning agents, and process for soldering.
Cleaning.
General requirements for securing of parts.
Cleaning and process of welding.

## 4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).

(b) Acceptance inspection.

(1) Acceptance inspection of equipment before preparation for delivery. (See 4.4.)

(2) Acceptance inspection of preparation for delivery. (See 4.19.)

4.3 Preproduction inspection.- This inspection will be performed by the Government, unless otherwise specified in the contract. It consists of (1) the preproduction inspection specified in Table III, (2) the inspection specified in the subsidiary documents covering the items listed in 4.4.1, and (3) the Group A and Group B inspections listed in Tables IV and V, respectively. Preproduction inspection will normally be performed in the following order: (1) Vibration, (2) Bounce. Other preproduction inspection may precede, follow, or, be interspersed among the above.

Table III.- Preproduction inspection

| Inspection Item | Req. Para. | Insp. Para. |
|---|---|---|
| Temperature | 3.9.1 | 4.10 |
| Altitude | 3.9.2 | 4.11 |
| Moisture | 3.9.3 | 4.12 |
| Rain | 3.9.4 | 4.13 |
| Sand and Dust | 3.9.5 | 4.15 |
| Vibration | 3.9.6 | 4.16 |
| Bounce | 3.9.7 | 4.17 |

4.4 Acceptance inspection of equipment before preparation for delivery.- The contractor, to demonstrate compliance with specified requirements, shall perform the inspection specified in 4.4.1, 4.5, and 4.6. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government, at its discretion, may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.5). Each unit, which will be subjected to group A, or group B inspection, shall be preconditioned after final assembly. (See 3.8) Test equipment for Government verification inspection shall be made available by the contractor.



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4.4.1 Acceptance inspection covered by subsidiary documents.- The following shall be inspected as provided by the applicable subsidiary documents. The inspection constitutes a part of the overall acceptance inspection required.

| Inspection Item | Para. | Inspection Item | Para. |
|---|---|---|---|
| Transformers | 3.4.1 | Fuses | 3.4.9 |
| Meters | 3.4.3 | Knobs | 3.4.10 |
| Connectors | 3.4.4 | Marking | 3.5 |
| Cable Assemblies | 3.4.5 | Symbols | 3.5.1 |
| Resistors | 3.4.6 | Instruction Plate | 3.5.2 |
| Switches | 3.4.7 | Panel Layout | 3.5.3 |
| Potentiometers | 3.4.8 | Finish | 3.6 |

4.5 Group A inspection.- This inspection, including sampling, shall conform to Table IV and the ordinary inspection procedures of Military Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government.

Table IV.- Group A inspection

| Inspection | Req. Para. | Insp. Para. | AQL Major | AQL Minor |
|---|---|---|---|---|
| Visual and Mechanical | | | 1.0 | 4.0 |
| Weight & Dimensions | 3.4.11 | 4.8 | | |
| Electrical Performance | | | | |
| Breakdown voltage | 3.7.1 | 4.9.1) | | |
| Power supply | 3.7.2 | 4.9.2) | 1.0 | ** |
| Current & voltage | 3.7.3 | 4.9.3) | for the | |
| Transient overload | 3.4.2.1 | 4.9.4) | group | |
| Regulation | 3.4.2.2 | 4.9.5) | combined | |
| Output ripple | 3.4.2.3 | 4.9.6) | | |

* Defects per 100 units. ** All electrical defects are considered major.

4.6 Group B inspection.- This inspection, including sampling, shall conform to Table V and to the special procedure for small sample inspection of Military Standard MIL-STD-105, The reduced inspection procedure shall be R-1. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection, and on samples selected from units that have been subjected to and met Group A inspection. The inspection level for normal and tightened inspection shall be L-6, and for reduced inspection, L-4.



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Table V.- Group B inspection

| Inspection and Test | Req. Para. | Insp. Para. | A Q L |
|---|---|---|---|
| Shock | 3.9.8 | 4.18 | 10% |

4.6.1 Re-inspection.- Unless otherwise specified, sample units which have passed Group B inspection, may be accepted on contract, provided that they are re-subjected to and pass Group A inspection. All damage incurred as the result of test action shall be repaired.

4.7 Bounce preconditioning.- The unit shall be placed in its normal operating position on the table of the Package Tester, Type 1000-SC, as made by the L. A. B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.8 Visual and mechanical inspection.- The unit shall be examined for the defects listed in Military Standard MIL-STD-252. In addition, the unit shall be inspected for the defects listed in Table VI, as follow

Table VI.- Classification of visual and mechanical defects.

| Classification | Defect |
|---|---|
| Major | Cover does not fit snugly.<br>Catches do not close properly.<br>Plugs do not fit exactly.<br>Welding is not intact. |
| Minor | Part is not completely fastened.<br>Cleaning is not thorough. |



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4.9 Performance. - The unit shall be tested to determine compliance with 3.7.

4.9.1 Breakdown voltage. - Insulation throughout the unit shall be tested for resistance to voltage breakdown in accordance with the values listed in Table VII. While testing, disconnections to protect circuit elements shall be made where necessary.

Table VII. - Breakdown Voltages.

| Operating Values | Test Values |
|---|---|
| 0 to 30 | 400 |
| 30 to 125 | 500 |

Note: Test voltages shall be applied for a minimum of ten seconds, and shall be either direct current or peak alternating current which is approximately sinusoidal in waveform and not over 100 cycles per second.

4.9.2 Power supply. - The unit shall be connected to a direct-current source of power which is maintained at 24 ± 0.2 volts. The power requirements specified in 3.7.1 shall be met as indicated in Table I.

4.9.3 Current and voltage. - The electrical performance in terms of current and voltage as indicated in Table II, shall be determined by application of the voltages indicated in Table VIII, as follows:

4.9.4 Transient overload. - The wave-shape amplitude between the collector and the emitter of the transistors shall be measured with an oscilloscope and shall conform to the requirements of 3.4.2.1.

4.9.5 Regulation. - The voltage regulation shall be checked with a voltmeter.

4.9.6 Output ripple. - The output ripple requirement shall be checked by the use of an oscilloscope whose characteristics permit evaluation of this characteristic to within ± ten percent.

Table VIII. - Test Jig ● Values

| Switch Position | Value of Simulated Load Resistor in Ohms | Power Rating of Resistor in Watts | Voltage | Power Supply Current in Milliamperes |
|---|---|---|---|---|
| A1 | 15 ± 5% | 1 | 1.4 ± 1% | 150 (Min) |
| A | 47 ± 5% | 2 | 6.6 ± 1% | 180 (Min) |
| B | 5600 ± 5% | 5 | 115.0 ± 1% | 34 (Min) |
| | Simulated Vehicle Power Supply | | 24.0 ± 1% | 800 (Min) |

● The term "test jig" refers to the simulated load resistors and the applied voltages listed in Table VIII.



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4.10 Temperature.- The unit shall be subjected to the temperature cycle in Military Standard MIL-STD-169. Measurements shall be taken at steps 1, 3, 5, 8, and 10. Performance shall conform to 3.7.

4.11 Altitude.- The unit, placed in its normal operating position in an altitude chamber, shall be tested as follows: (Note: To check temperature stabilization, place a thermocouple on the largest internal mass centrally located within the unit.)

(a) With the unit placed in the chamber, operating and stabilized at room temperature, take the following measurements:

| Paragraph | Measurement |
|---|---|
| 4.9.3 | Current and voltage (Any one switch position) |

(b) Reduce the barometric pressure to 20.6 ± 0.1 inches of mercury (simulated 10,000 feet above sea level) and, after temperature and pressure stabilization, repeat the measurements specified in 4.11(a). The unit shall meet specified performance for those measurements.

(c) Remove the power from the unit and further reduce the barometric pressure to 3.4 ± 0.1 inches of mercury (simulated 50,000 feet above sea level.) Maintain this condition for 1 hour.

(d) Return the chamber to ambient pressure. Apply power to the unit and, after stabilization, repeat the measurements specified in 4.11(a). The unit shall meet specified performance for these measurements.

4.12 Moisture resistance.- The unit shall be subjected to humidity cycling per Military Standard MIL-STD-170. The procedure shall be as follows:

(a) Dry the unit at 130 ± 5 degrees Fahrenheit for 24 hours.

(b) Condition at a temperature of 77 ± 5 degrees F. and with a relative humidity of 40 to 50% for 24 hours.

(c) Take measurements as specified in 4.12.1.

(d) Subject the unit to five 48 hour cycles consecutively, per MIL-STD-170. Take measurements as shown on the standard as specified in 4.12.1. The requirements shall be met as specified.

(e) After completion of all cycling, condition the unit at 77 ± 5 degrees F. for 24 hours, with a relative humidity of 40 to 60%.

(f) Take measurements as specified in 4.12.1. The requirements shall be met as specified.



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4.12.1 Performance.- The unit shall meet the performance specified in Table II.

4.12.2 Provisions.-

(a) The unit shall not be removed from the test chamber in order to take measurements.

(b) Measurements shall be initiated within five minutes after the application of power and be completed witnin as short a period of time as possible. Power shall be shut off when all measurements have been completed.

4.13 Rain.- The unit shall be tested as follows:

(a) Dry the unit at 150 ± 5 degrees F. for 48 hours.

(b) Maintain the unit for four hours at 77 ± 5 degrees F. and a relative humidity of from 40 to 60 percent.

(c) Take measurements as specified in Table II to insure that the unit meets the specified performance.

(d) Place the unit in its normal operating position in a test chamber maintained at 77 ± 5 degrees F. Allow simulated rain to fall on exposed surfaces of the unit. The water shall be at a temperature of 65 ± 5 degrees and fall freely at a rate of two inches an hour. Continue this for one hour.

(e) With rain at 2 inches per hour introduce a simulated wind blowing horizontally at 20 ± 5 miles per hour, against one side of the equipment. The rain shall fall on exposed surfaces and be blown on one side of the unit. This condition shall be maintained for one hour.

(f) Turn the unit in a horizontal plane approximately 90 degrees at intervals of one hour and expose all four sides to the wind and rain continuously.

4.14 Performance check.- After the above specified exposure, check the unit for conformance with 3.7. This check shall be completed within an hour after completion of the exposure and shall establish that specified requirements have been maintained. Furthermore, no water shall have entered the unit.



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4.15 Sand and dust.- The unit shall be tested as follows:

(a) Place the unit in its normal operating position, in the test chamber. Raise and maintain the sand and dust density at 0.1 to 0.5 gram per cubic foot within the test space. The relative humidity shall not exceed 30 percent at any time during the test.

(b) The sand and dust shall be of angular structure and shall have a particle size such that the particles can pass through screens as follows:

| Screen, U. S. Standard Sieve Series | Percent |
|---|---|
| 100 - mesh | 100 |
| 140 - mesh | 98 ± 2 |
| 200 - mesh | 90 ± 2 |
| 325 - mesh | 75 ± 2 |

(c) Chemical analysis of the sand and dust shall be as follows:

| Substance | Percent by weight |
|---|---|
| $SiO_2$ | 97 to 99 |
| $Fe_2O_3$ | 0 to 2 |
| $Al_2O_3$ | 0 to 1 |
| $TiO_2$ | 0 to 2 |
| MgO | 0 to 1 |
| Ignition losses | 0 to 2 |

(d) Maintain the internal temperature of the test chamber at 77 ° ± 4 ° F for a period of 6 hours, with the sand and dust velocity through the test chamber between 100 and 500 feet per minute.

(e) After 6 hours at the above conditions, raise the temperature to 160 ° ± 4 ° F and maintain it at this value for 6 hours. The other conditions shall remain the same.

(f) Remove the unit from the test chamber and allow it to cool to room temperature. Accumulated dust shall be removed by brushing, wiping, or shaking, with care taken to avoid introduction of additional dust. However, dust shall not be removed by either blast or vacuum cleaning. The unit shall meet specified performance for the following measurements.

| Paragraph | Measurement |
|---|---|
| 3.7 | 4.9 |

(g) Open the unit. There shall be no evidence of deterioration of either external or internal parts.



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4.16 Vibration.- Internal vibration of the unit shall be determined as follows:

(a) Secure the unit directly to a vibration table that can be controlled to within 10 percent of the specified amplitude. The mounting method shall be such that vibration within the unit can be observed and measured. To facilitate this observation and measurement, subassemblies may be tested separately, provided they are secured to the table in a manner similar to that used to mount them in the unit.

(b) Vibrate the unit successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excursions of the applied vibration shall be not less than 0.030 inch.

(c) In each of the three directions, change the frequency in steps of one cycle per second, and maintain each frequency for at least 10 seconds.

(d) Measure vibration amplitudes by optical means, or by other means, provided that vibration of the part is not affected by the technique employed.

4.17 Bounce.- The unit shall be tested on the package tester, type 1000-SC, as made by the L. A. B. Corporation, Skaneateles, New York, or equal. Accessories shall be selected from those listed on Drawing SC-A-46439. The test shall be as follows:

(a) Cover the tester bed with a panel of 1/2-inch plywood, with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the surface, at 6-inch intervals around all four edges, and at 3-inch intervals, in a 6-inch square in the center.

(b) Place the unit , prepared for field transport, on the bed of the package tester. Limit the lateral motion, by wooden fences, to not more than 3 inches and not less than 1 inch. Additional barriers may be used to prevent tumbling, provided that the fore-and-aft motion of the equipment against the backstop is not restrained.

(c) Operate the package tester, shafts in phase, for a total of 3 hours at 284 ± 2rpm. Turn the unit at the end of each 30 minutes so that it will rest on a new face.

4.18 Shock, Bench-handling.- The chassis and front panel assembly shall be removed from its enclosure, as for servicing, and placed in a suitable position for servicing on a solid, 2-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

(a) Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axes, for a total of four drops.

(b) Repeat "(a)" with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing.



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4.19 Acceptance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116.

PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A.- The unit (TS-1348( )/U) shall be preserved, packaged and tested in accordance with the applicable provisions of Specification MIL-P-116 and in a manner that will afford adequate protection against corrosion, deterioration and damage during worldwide shipment, handling and open storage.

5.1.2 Level C.- The unit shall be afforded preservation and packaging in accordance with the suppliers' normal commercial practice.

5.2 Packing.- The unit packaged as specified, shall be packed in shipping containers conforming to the requirements of the specifications referenced below for the designated level. The gross weight shall not exceed the weight shown unless an individually packed item exceeds that amount. Closure and strapping shall be as prescribed in the applicable box specification or appendix thereto, except that bands of reinforced, pressure-sensitive adhesive tape, 1/2 inch wide and conforming to Specification PPP-T-97 for Grade IV, shall be used for fiberboard boxes in lieu of metal strapping.

5.2.1 Level A.-

| BOX SPECIFICATION | | MAX. GROSS WT. |
|---|---|---|
| Fiberboard (Type I, Class 2) | PPP-B-636 | Table VI |
| Nailed Wood (Style 4, Cl 2) | PPP-B-621 | 200 lbs. |
| Wirebound (Style 2 or 3, Class 2) | PPP-B-585 | 200 lbs. |
| Wood Cleated Fiberboard (Overseas) | PPP-B-591 | 200 lbs. |
| Wood Cleated Plywood (Overseas) | PPP-B-601 | 200 lbs. |

5.2.2 Level B.-

| BOX SPECIFICATION | | MAX. GROSS WT. |
|---|---|---|
| Fiberboard (Type I, Class 1) | PPP-B-636 | Table I |
| Nailed Wood (Style 4, Class 1) | PPP-B-621 | 200 lbs. |
| Wirebound (Style 3, Class 1) | PPP-B-585 | 200 lbs. |
| Wood Cleated Fiberboard (Domestic) | PPP-B-591 | 200 lbs. |
| Wood Cleated Plywood (Domestic) | PPP-B-601 | 200 lbs. |



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5.2.3 Level C.- The unit shall be packed in shipping containers of the type, size and kind commonly used for the purpose in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the rules or regulations of the common carrier, applicable to the mode of transportation.

5.2.4 Pilot Pack.- When Level "A" packaging and packing is specified above, one acceptable model of the unit will be pilot packed in a manner that will pass the preproduction tests specified in Specification MIL-P-116. Illustrations of the packaging and packing procedure, together with Bills of Material, will be prepared and furnished in accordance with the provisions of Signal Corps Drawing No. SC-A-362100.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

6 NOTES

6.1 Intended use.- This unit is designed to test radiosondes prior to their release in flight. Means are provided to measure operating currents and voltages developed in the radiosonde, when powered with the usual battery, or when powered by an external source such as the battery power available from vehicular transport units. Both Radiosondes AN/AMT-4 and AN/AMT-12 may be tested. Means are also provided to check the serviceability of the Battery Pack BA-259/AM.

6.2 Ordering data.- Procurement should specify the following:

(a) The title, number, and date of this specification, and any amendment thereto.

(b) The levels of packaging and packing required.

(c) The particular paragraphs of Section 5 which are applicable to the procurement.

(d) Marking and shipping of samples.

(e) Place of final inspection.

(f) Technical literature required. (See 3.10).



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6.2 Ordering data.- (Cont'd)

(g) Quantity of running spares. (See 3.10.)

(h) Provide for the submission of both the sample of the Instruction Plate (see 3.5.2), and for submission of the detailed drawing of the Panel Layout and Marking (see 3.5.3.)

6.3 Color.- The color chip furnished by the procuring agency will be No. X-24087, and may be obtained by request from the Commanding General, ATTN: SIGSU-J4b, U. S. Army Signal Supply Agency, 225 South Eighteenth Street, Philadelphia 3, Pennsylvania.

6.4 Nomenclature.- The parenthesis in the nomenclature will be replaced by a letter identifying the particular design; for example, TS-1348W/GMM-1. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1).

6.5 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) Type A-- That portion of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Military Standard 105.

(b) Type B-- That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Military Standard 105.

(c) Type C-- A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United State Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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